How similar are the treatment responses to biosimilars in patients with psoriasis? A systematic review of statistical margins in comparative clinical trials

To the Editor: Biosimilars are medications that have no clinically meaningful differences in terms of safety, purity, and potency relative to a reference drug product that has already obtained approval from the Food and Drug Administration.[1] These drugs are often investigated by equivalence trials wherein the biosimilar is determined clinically to be neither inferior nor superior when compared with its originator. Because of the complex molecular nature of biosimilars, manufacturing identical replicas of corresponding originators is impossible.[2-4] Thus, it is crucial to decide upon the acceptable margin of error (MoE), which is defined as the maximum allowable difference in a standard treatment outcome measure between the biosimilar and originator. Therefore, it is critical to understand the MoE in studies on biosimilars to determine if potentially clinically important differences in efficacy were ruled out statistically by study design.

We conducted a systematic review using MEDLINE, Embase, Cochrane Central Register of Controlled Trials, LILACS, and ClinicalTrials.gov from inception through February 17, 2017. To obtain data from unpublished or unidentified studies, we searched clinicaltrialresults.org and contacted corresponding authors. Our inclusion criteria included patients ≥18 years of age with plaque psoriasis randomized to receive the biosimilar or originator with measured Psoriasis Area and Severity Index (PASI) endpoints at week 12 or later.

We abstracted the following data: study design, sample size, study power, confidence intervals, MoE, outcomes, and response rates. Four clinical trials that included 1746 adult patients qualified for inclusion; the MoEs in these reports ranged from ±14% to ±18% (Fig 1, Table I). Five clinical trials were excluded because the information available was not sufficient, and industry sponsors declined to provide this information upon request.

These margins, which were decided upon by regulatory bodies, might be unacceptably wide (eg, imprecise) to some prescribing clinicians. For example, if the true PASI 75 response rates were 65% and 50% for the originator and biosimilar, respectively, this would be considered equivalent responses in a study with ±15% MoE. One physician might consider these parameters sufficient and comfortably prescribe the biosimilar as a replacement for the originator; however, another clinician might require more conservative margins. Moreover, this quandary is compounded by factors that influence the MoE such as the study's sample size. For example, a trial powered at 0.80 with a ±15% MoE would require, based on typical statistical specifications needed in clinical trials, 382 patients. However, the identical study with the same parameters yet mandating a ±10% MoE would require an additional 476 patients (n = 858). Other aspects of a study, such as the variability in the study population, expertise, and geographic location of the study sites, are critical to understand, yet are not understood by the average practitioner.

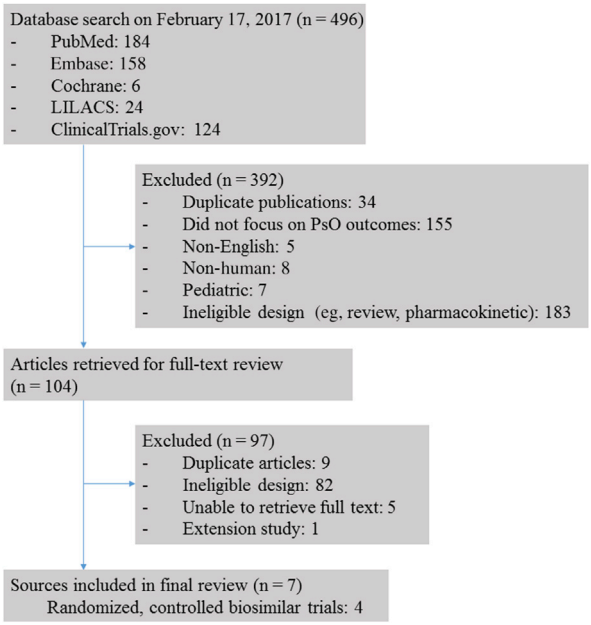

Fig 1. Evidence search and selection. *PsO*, Psoriasis.

As biosimilar approvals accrue and their use increases, possibly delivering decreased healthcare costs and better patient access, practitioners require a deeper understanding of the design and conduct of clinical trials to better interpret the true likelihood of achieving the same clinical outcome regardless of drug choice.

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Table I. Characteristics of the double-blinded randomized, controlled trials comparing biosimilars with the originators

Source (est. study completion year)	Trial name/ registry no.	Biosimilar vs originator	Study design	Study regimen	Primary outcome (wk measured)	No. of total participants (no. biosimilar/no. originator)	Response rate, %	CI, %	Prespecified upper/lower margin, %	Study power
Amgen PUI: L72275951 (2016)	Study 263/ NCT01970488	Adalimumab-atto (AMJEVITA) vs adalimumab (HUMIRA)	Period 1 (wks 1-16): ABP-501 or EU-approved HUMIRA. Period 2 (wks 16-52): subjects with PASI 50 continue study. ABP-501 subjects continue treatment with ABP-501. EU-approved HUMIRA subjects are randomized 1:1 to transition to ABP-501 or continue EU-approved HUMIRA.	80 mg at wk 1 and 40 mg at wk 2 and every other wk thereafter	Mean percent improvement in PASI from baseline (16)*	350 (175/175)	86.6 vs 88	90	15/−15	0.90
Sandoz PMID: 27787890 (2015)	Study 302/ NCT01891864	Etanercept-szzs (Erelzi) vs etanercept (ENBREL)	Period 1 (wks 1-12): etanercept or GP2015. Period 2 (wks 12-30): subjects achieving PASI 50 in each group randomized to continue initial treatment or undergo three treatment switches at 6-wk intervals. Extension period (wks 30-52): maintenance of last assigned drug.	50 mg SC 2x/wk for the first 12 wks and 50 mg 1x/wk thereafter	PASI 75 (12)	531 (264/267)	70.4 vs 71.6	90	18/−18	0.90
Coherus BioSciences PUI: L613888021 (2016)	NCT02134210	CHS-0214 vs etanercept (ENBREL)	Subjects were randomized to CHS-0214 or etanercept.	50 mg SC 2x/wk for the first 12 wks and then 4x/wk for 36 wks	PASI 75 (12)	521 (261/260)	64.5 vs 62.3	95	18/−18	TBD

BIOCAD (2016)	NCT02762955	BCD-057 vs adalimumab (HUMIRA)	Subjects were randomized to BCD-057 or adalimumab. Subjects in adalimumab group undergo crossover in 1:1 ratio after wk 24.	80 mg loading dose at wk 0, then 40 mg SC every other wk through wk 52	PASI 75 (16)	344	TBD	95	14/−14	0.80

There were insufficient data from studies NCT02016105 (Sandoz), NCT02850965 (Boehringer Ingelheim), NCT02660580 (EMD Serono), NCT02714322 (Mylan), and NCT02581345 (Baxalta US) because sponsors declined to provide the information upon request.
CI, Confidence interval; *est.*, estimated; *PASI*, Psoriasis Area and Severity Index; *PASI 50*, 50% reduction in Psoriasis Area and Severity Index score; *PASI 75*, 75% reduction in Psoriasis Area and Severity Index score; *PMID*, PubMed identification number; *PUI*, Embase identification number; *SC*, subcutaneous; *TBD*, to be determined.
*Percent improvement in mean and median PASI score from baseline measurement.

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Prevalence of chronic hepatitis B and C in psoriasis patients: A cross-sectional study in a large US population

To the Editor: An association between psoriasis and hepatitis B and C is inconsistently reported in the literature.[1-3] Because both hepatitis B and C viruses have been hypothesized to exacerbate psoriasis and because biologic treatments for psoriasis may be contraindicated in patients infected with hepatitis viruses, exploring this possible relationship for each virus may lead to better management.[1] Consequently, this study aimed to determine the prevalence of hepatitis B and C in patients with psoriasis within a large, single-center, US patient population.

Under an Institutional Review Board—approved protocol, we utilized a large, urban, electronic medical record data repository, Northwestern Medicine Enterprise Data Warehouse (NMEDW) with data on more than 4 million individuals.[4] A very high percentage of the patients in the NMEDW data repository are from the Chicago metropolitan area. A very small minority of patients are from surrounding states and/or rural areas. All patients, regardless of billing status, are included. We searched the NMEDW for all patients aged 18 to 89 years who underwent hepatitis B and/or C serology screening between September 2010 and September 2016. We then divided this group into those with a diagnosis of psoriasis (*International Classification of Diseases, 9th Revision* and *International Classification of Diseases, 10th Revision* codes: 696.1, L40.0-L40.4, L40.8, and L40.9) and a control cohort consisting of all patients aged 18 to 89 years within the same time frame but without a diagnosis of psoriasis. The main outcome measure was a concurrent diagnosis of hepatitis B and/or C (*International Classification of Diseases, 9th Revision* and *International Classification of Diseases, 10th Revision* codes: B18.0, B18.1, B18.2, B19.10, B19.11, B19.20, B19.21, B19.9, 070.2, 070.3, 070.4, 070.5, and 070.7). Logistic regression analysis was performed to obtain odds ratios (ORs) with 95% confidence intervals (CIs) and corresponding *P* values.

Table I. Demographic characteristics

Demographics	Population with psoriasis n (%)	Population without psoriasis n (%)	*P* value
Total	2590	112,265	
Sex			<.001
Female	1287 (49.7)	64,854 (57.8)	
Male	1303 (50.3)	47,346 (42.2)	
Not specified	—	65 (0.1)	
Race			<.001
White	1552 (59.9)	55,544 (49.5)	
Black	184 (7.1)	16,654 (14.8)	
Other	399 (15.4)	20,170 (18)	
n/a	455 (17.6)	19,897 (17.7)	
Mean age (±SD), y	49.1 (±14.6)	45.4 (±15.2)	<.001

n/a, Not available; *SD*, standard deviation.

A total of 114,855 patients underwent hepatitis B and/or C screening (Table I). The frequency of hepatitis B and/or C is shown in Table II. There was no significant difference between the populations with and without psoriasis in terms of prevalence of hepatitis B (OR, 0.56; 95% CI, 0.34-1.03; $P = .064$) or hepatitis C (OR, 0.8; 95% CI, 0.57-1.13; $P = .2$).

Using a cross-sectional design to analyze a large, urban, single-center, midwestern US patient population, this study found no significant difference in prevalence for chronic hepatitis B or C in patients with psoriasis compared with in those without psoriasis. Limitations of this study include the cross-sectional design, the lack of adjustment for additional risk factors for hepatitis B and C, and the inability to assess prevalence of hepatitis B and/or C in patients with psoriasis who were not screened for hepatitis. Additionally, prevalence might have been affected by differences in demographics between psoriasis and control populations. Moreover, this study, although informed by data on a large number of patients, was conducted in a single, urban, multiethnic, academic-based medical center and is therefore not viewed as representative of the general population. Strengths of this study include its large